Curriculum Vitae - Asikul Islam

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EDUCATION

Doctor of Philosophy, Mathematics **Expected May 2027**
University of Houston **Houston, TX, USA**

- Dissertation (In Progress): "*Tensorial Reduced-Order Models for Parameter Estimation in Brain Tumor Growth Models*"
  Advisor: Dr. Andreas Mang
- Developed a solver for single- and multi-species reaction–diffusion models of brain tumor growth on realistic 2D and 3D brain geometries (finite volume discretization, fictitious domain method, second-order Strang splitting with Crank–Nicolson and Runge–Kutta steps).
- Designed and implemented tensorial reduced-order models (HOSVD/Tucker and Tensor Train, intrusive and non-intrusive) for a six-species, nine-parameter model; achieved speedups of 85 to 120 over the full-order solver.
- Formulated parameter estimation as a PDE-constrained optimization problem; derived discrete adjoints, gradients, and reduced Hessians (discretize-then-optimize) and implemented and compared L-BFGS, Gauss–Newton, Newton, and log-barrier trust-region methods with Tikhonov regularization and the Morozov discrepancy principle.
- Contributed to a gradient-consistent reduced-order inversion with parameter-dependent Tucker bases, and correction of ROM-induced parameter error via hyper-differential sensitivity analysis.

Master of Science, Applied Mathematics **Fall 2025**
University of Houston **Houston, TX, USA**

- Conducted research on Efficient Numerical Methods for Multispecies Tumor Growth Simulations.

Master of Science, Mathematics **December 2017**
University of Dhaka **Dhaka, Bangladesh**

- Conducted research on applications of fuzzy logic in business, finance, and management.

Bachelor of Science, Mathematics **August 2015**
University of Dhaka **Dhaka, Bangladesh**

- Conducted a thesis on tensor flow and tensor analysis.

WORK EXPERIENCE

Graduate Research Assistant (NSF-funded (DMS-2145845)) **August 2024 – July 26**
University of Houston **Houston, TX, USA**

- Wrote and maintain the group's MATLAB/Python code for tumor growth simulation, reduced-order modeling, and inversion.
- Ran large multi-parameter simulation studies on the University of Houston HPC cluster; built a memory-efficient offline pipeline (spatial pre-compression, randomized SVD, out-of-core snapshot processing).
- Co-authored one manuscript under review and two in preparation; presented results at the SIAM Annual Meeting 2026 and in several poster sessions.

Graduate Teaching Assistant **August 2022 - Present**
University of Houston **Houston, TX, USA**

- Lead Teaching Assistant for Statistics for the Sciences (MATH 3339), coordinate the course's teaching assistants.
- Prepare homework solution sets and organize grading for midterm and final exams.
- Hold office hours, review sessions, and one-on-one and group tutoring.

Lecturer **January 2022 - July 2022**
Northern University of Bangladesh **Dhaka, Bangladesh**

- Taught undergraduate Statistics (probability distribution, statistical inference, hypothesis testing) and Calculus I and II for engineering and science students.

Lecturer **January 2018 - December 2021**
Canadian University of Bangladesh **Dhaka, Bangladesh**

- Taught six undergraduate courses with class sizes of 70-100: Differential and Integral Calculus, Probability and Statistics, Fourier and Laplace Transform, Numerical Analysis, Engineering Math and Complex Variables (over 1000 students in total).
- Wrote lesson plans and exams, ran review sessions and workshops and contributed to curriculum.

RESEARCH INTEREST

- Reduced-order modeling for nonlinear and parametric partial differential equations.
- Low-rank approximation for nonlinear parametric equations.
- Tensor-based reduced order models.
- Large-scale scientific computing and efficient simulation of complex systems.
- PDE-constrained optimization.
- Inverse problems and uncertainty quantification.
- Scientific machine learning.

RESEARCH EXPERIENCE

- Tensorial reduced order models and PDE-constrained optimization.
- Application of Fuzzy Logic to explore the techniques for forecasting, decision making and control based on "*if…then*" rules in environments characterized by uncertainty, vagueness and imprecision.
- A comparative study of various shortest path algorithms.

PUBLICATIONS

- Asikul Islam, Md Rezwan Bin Mizan, Maxim Olshanskii and Andreas Mang (2026), Tensorial Reduced-Order Models for Parametric Coupled Reaction-Diffusion Systems: Application to Brain Tumor Growth Modeling, arXiv preprint arXiv:2603.14101. (Under Review).

MANUSCRIPTS IN PREPARATION

- A. Islam, A. Pegah, M. Olshanskii, and A. Mang: Discretize-then-optimize Parameter Estimation for Brain Tumor Growth Models using Tensorial ROMs.
- A. Pegah, A. Islam, and A. Mang: Correcting Reduced-Order Model Error in Tumor Growth Inversion via Hyper-Differential Sensitivity Analysis.

SKILLS

Programming Languages	MATLAB, Python, FEniCSx, Fortran, Mathematica.
Numerical Methods	Finite difference and finite volume methods; operator splitting (Strang, Crank–Nicolson, Runge–Kutta); proper orthogonal decomposition (POD); tensor decompositions (HOSVD/Tucker, Tensor Train); randomized SVD.
Optimization and Inverse Problems	Discrete adjoints, gradients, and Hessians; L-BFGS, Gauss–Newton, Newton, and trust-region methods; Tikhonov regularization; Morozov discrepancy principle.
Computing	HPC cluster computing; out-of-core processing of large snapshot data; LaTeX; Git.

PROFESSIONAL & CAMPUS INVOLVEMENT

- Volunteered at SIAM Annual Meeting 2026, Cleveland.
- Volunteered at CBMS Conference: Research at the Interface of Applied Mathematics and Machine Learning 2025, University of Houston.
- Volunteered at ChAMELEON Summer School 2025, University of Houston.
- Volunteered at SIAM Texas-Louisiana Sectional Meeting 2022 at the University of Houston.

AWARDS

2025 Best Poster Award from the SIAM Texas-Louisiana Sectional Meeting at the University of Texas at Austin, TX, US.

CONFERENCES, WORKSHOPS AND SEMINARS

Professional Development

2026 Bayesian Inverse Problems and UQ Workshop, by Institute for Computational and Experimental Research in Mathematics (ICERM), Brown University.
2025 CBMS Conference: Research at the Interface of Applied Mathematics and Machine Learning 2025, University of Houston.
2025 SIAM Texas-Louisiana Sectional Meeting, UT Austin, TX, US.
2025 ChAMELEON Summer School 2025, University of Houston.
2025 Scientific Machine Learning Summer School, Texas A&M Institute of Data Science.
20 Workshop on 'Firedrake 25', Department of Mathematics, Baylor University.
2022 SIAM Conference Texas-Louisiana Section, University of Houston, TX, US.
2019 Webinar on "Faculty Development Program on OBE", Canadian University of Bangladesh.
2019 Workshop on "Structural Equation Modeling (SEM) and writing academic research paper", Canadian University of Bangladesh.
2018 Self-Development workshop on "Goal setup, presentation, volunteerism and first step into professional development" organized by the Life Carnival Foundation.

Contributed Conference Presentations

2026 Talk: "*Fast Tensorial Surrogate Models for Parametric Brain Tumor Growth Dynamics in Many-Query Regimes*". SIAM Annual Metting 2026, Cleveland.
2025 Poster Presentation at the CBMS Conference: Research at the Interface of Applied Mathematics and Machine Learning, University of Houston.
2025 Poster Presentation at the SIAM Conference Texas-Louisiana Section, UT Austin, TX, US.
2025 Poster Presentation at the ChAMELEON Summer School 2025, University of Houston.

Honors & Fellowships

- Graduate Tuition Fellowships — Aug 22- Present
  University of Houston
- Teaching Assistantship — Aug 22-Present
  University of Houston

MEMBERSHIPS

- American Mathematical Society (AMS) — Jan 23-Present
- Society for Industrial and Applied Mathematics (SIAM) — Aug 23-Present
- Bangladesh Mathematics Society (BMS) — Aug 17-Present

RELEVANT GRADUATE COURSES TAKEN

Optimization | Computational Methods | Numerical PDE | Differential Equations | Advanced Linear Algebra | Numerical Analysis | Statistics | Complex Variables

REFERENCE

Will be provided upon request.